

## JERUSALEM CENTER, Israel

WMO: 401830

Lat: 31.7806N

Lon: 35.2217E

Elev: 810

StdP: 91.97

Time Zone: 2.00 (ISR)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.8        | 4.3 | -12.4                         | 1.4 | 17.0 | -9.7 | 1.8 | 16.0 | 10.2                  | 6.9  | 9.2 | 6.3  | 4.4                   | 260  | 0.390 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.3                    | 33.6            | 18.0 | 32.3 | 18.0 | 31.1 | 18.0 | 21.9                | 28.8 | 21.2 | 27.8 | 20.6 | 26.9 | 3.3                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.0                            | 16.2 | 24.0 | 19.4 | 15.6 | 23.5 | 18.8 | 15.0 | 23.0 | 68.3          | 28.8 | 65.6 | 27.9 | 63.4 | 26.9 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 7.8               | 6.8  | 6.1 | DB                         | 1.3  | 36.5               | 1.6 | 1.7                                                | 0.2  | 37.7       | -0.7 | 38.8       | -1.6 | 39.7       | -2.7 | 41.0 |  |
|                   |      |     | WB                         | -0.4 | 23.2               | 1.4 | 0.9                                                | -1.4 | 23.9       | -2.2 | 24.4       | -3.0 | 24.9       | -4.0 | 25.6 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.6   | 10.0  | 11.2  | 14.4  | 17.8  | 21.0  | 23.5  | 25.2  | 25.5  | 24.0  | 21.4  | 16.7  | 12.2  |
|                                                                                  | DBStd       | 6.33   | 3.24  | 3.80  | 4.48  | 4.34  | 3.97  | 2.86  | 2.09  | 1.96  | 2.25  | 3.20  | 3.51  | 3.04  |
|                                                                                  | HDD10.0     | 83     | 38    | 26    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 11    |
|                                                                                  | HDD18.3     | 952    | 258   | 201   | 140   | 64    | 17    | 0     | 0     | 0     | 0     | 7     | 72    | 192   |
|                                                                                  | CDD10.0     | 3226   | 38    | 60    | 144   | 233   | 342   | 406   | 471   | 480   | 419   | 353   | 202   | 79    |
|                                                                                  | CDD18.3     | 1053   | 1     | 2     | 18    | 46    | 101   | 157   | 212   | 222   | 169   | 101   | 23    | 2     |
|                                                                                  | CDH23.3     | 8011   | 0     | 3     | 110   | 318   | 850   | 1320  | 1816  | 1822  | 1138  | 565   | 67    | 1     |
|                                                                                  | CDH26.7     | 2661   | 0     | 0     | 22    | 78    | 281   | 459   | 656   | 662   | 347   | 149   | 6     | 0     |
| Wind                                                                             | WSAvg       | 3.2    | 3.1   | 3.1   | 3.2   | 3.1   | 3.3   | 3.6   | 3.8   | 3.6   | 3.3   | 2.8   | 2.6   | 2.8   |
| Precipitation                                                                    | PrecAvg     | 590    | 133   | 118   | 93    | 30    | 4     | 1     | 0     | 0     | 0     | 19    | 68    | 135   |
|                                                                                  | PrecMax     | 1153   | 422   | 353   | 242   | 220   | 30    | 28    | 0     | 0     | 3     | 81    | 257   | 476   |
|                                                                                  | PrecMin     | 280    | 14    | 19    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 7     |
|                                                                                  | PrecStd     | 194    | 82    | 79    | 47    | 41    | 7     | 5     | 0     | 0     | 1     | 20    | 59    | 106   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 21.0  | 23.4  | 29.4  | 31.4  | 34.5  | 34.7  | 35.0  | 35.6  | 33.5  | 32.8  | 27.7  | 22.6  |
|                                                                                  |             | MCWB   | 10.5  | 11.8  | 13.9  | 13.4  | 16.2  | 16.8  | 20.0  | 19.7  | 18.5  | 15.8  | 14.1  | 10.5  |
|                                                                                  | 2%          | DB     | 17.8  | 21.3  | 26.3  | 28.8  | 32.0  | 32.7  | 32.8  | 32.7  | 31.7  | 30.2  | 25.3  | 20.1  |
|                                                                                  |             | MCWB   | 9.2   | 10.2  | 12.8  | 12.9  | 15.2  | 17.3  | 18.9  | 20.0  | 18.9  | 16.4  | 13.2  | 10.2  |
|                                                                                  | 5%          | DB     | 15.8  | 19.2  | 23.7  | 26.8  | 29.8  | 30.8  | 31.3  | 31.2  | 30.1  | 28.3  | 23.3  | 18.1  |
|                                                                                  |             | MCWB   | 8.5   | 9.4   | 11.4  | 12.5  | 14.8  | 16.8  | 18.8  | 19.5  | 18.8  | 15.9  | 12.5  | 8.9   |
|                                                                                  | 10%         | DB     | 14.5  | 17.1  | 21.5  | 24.8  | 27.9  | 29.3  | 30.0  | 30.0  | 28.7  | 26.6  | 21.7  | 16.4  |
|                                                                                  |             | MCWB   | 8.0   | 8.6   | 10.8  | 12.3  | 14.3  | 16.8  | 18.7  | 19.5  | 18.3  | 15.7  | 12.3  | 9.1   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 11.9  | 13.4  | 15.6  | 16.1  | 18.7  | 21.1  | 22.8  | 23.2  | 22.6  | 20.5  | 17.0  | 13.6  |
|                                                                                  |             | MCDB   | 16.5  | 21.2  | 25.8  | 23.9  | 28.6  | 29.9  | 30.2  | 31.3  | 29.0  | 25.7  | 23.4  | 18.5  |
|                                                                                  | 2%          | WB     | 10.7  | 11.6  | 14.0  | 14.8  | 17.2  | 19.9  | 21.5  | 22.1  | 21.4  | 19.2  | 15.4  | 12.4  |
|                                                                                  |             | MCDB   | 14.7  | 17.5  | 22.2  | 22.8  | 26.5  | 28.1  | 28.7  | 29.1  | 27.3  | 24.8  | 20.7  | 16.3  |
|                                                                                  | 5%          | WB     | 9.9   | 10.6  | 12.7  | 13.9  | 16.4  | 19.0  | 20.7  | 21.4  | 20.5  | 18.3  | 14.4  | 11.5  |
|                                                                                  |             | MCDB   | 13.5  | 16.1  | 20.5  | 23.0  | 25.6  | 26.6  | 27.7  | 27.9  | 26.2  | 23.4  | 19.4  | 14.7  |
|                                                                                  | 10%         | WB     | 9.2   | 9.8   | 11.7  | 13.1  | 15.6  | 18.2  | 20.1  | 20.8  | 19.9  | 17.5  | 13.7  | 10.6  |
|                                                                                  |             | MCDB   | 12.6  | 15.0  | 19.3  | 22.4  | 24.7  | 25.6  | 26.8  | 26.9  | 25.3  | 22.4  | 18.5  | 13.6  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.4   | 6.1   | 7.5   | 8.7   | 9.2   | 9.4   | 9.5   | 9.3   | 9.0   | 7.7   | 6.2   | 5.3   |
|                                                                                  |             | MCDBR  | 7.5   | 8.5   | 9.8   | 10.5  | 10.7  | 10.7  | 10.3  | 9.9   | 10.2  | 9.2   | 8.0   | 7.2   |
|                                                                                  |             | MCWBR  | 4.2   | 4.5   | 4.9   | 5.2   | 5.3   | 6.2   | 6.5   | 5.8   | 6.0   | 4.7   | 4.1   | 3.8   |
|                                                                                  | 5% WB       | MCDBR  | 6.3   | 7.5   | 8.8   | 9.7   | 9.8   | 9.9   | 10.1  | 9.9   | 9.7   | 8.5   | 7.0   | 5.7   |
|                                                                                  |             | MCWBR  | 4.0   | 4.4   | 5.2   | 5.8   | 6.0   | 5.9   | 5.9   | 5.2   | 5.4   | 4.7   | 3.9   | 3.8   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.345  | 0.374 | 0.415 | 0.452 | 0.449 | 0.385 | 0.394 | 0.401 | 0.395 | 0.409 | 0.362 | 0.344 |       |
|                                                                                  | taud        | 2.303  | 2.201 | 2.069 | 1.977 | 2.014 | 2.241 | 2.241 | 2.224 | 2.205 | 2.225 | 2.169 | 2.289 | 2.321 |
|                                                                                  | Ebn at Noon | 867    | 873   | 861   | 843   | 846   | 898   | 887   | 878   | 870   | 824   | 839   | 846   |       |
|                                                                                  | Edh at Noon | 106    | 129   | 159   | 181   | 176   | 140   | 142   | 142   | 134   | 132   | 107   | 99    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.05   | 3.87  | 5.26  | 6.44  | 7.47  | 8.24  | 8.03  | 7.37  | 6.28  | 4.77  | 3.56  | 2.86  |       |
|                                                                                  | RadStd      | 0.19   | 0.28  | 0.32  | 0.28  | 0.29  | 0.12  | 0.08  | 0.09  | 0.15  | 0.19  | 0.18  | 0.17  |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (18 neighbors) | +0.46 | N/A     | -1.99  | N/A     | N/A   | N/A   | N/A         | -62     | +151    | +99     |

Nomenclature: See separate page